

CURRICULUM VITAE

University of Idaho

NAME: Matthew R. Falcy

DATE: September 25, 2026

RANK OR TITLE: Associate Professor of Biometrics

DEPARTMENT: Fish and Wildlife Sciences

OFFICE LOCATION AND CAMPUS ZIP:

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DATE OF FIRST EMPLOYMENT AT UI: March 1, 2021

DATE OF TENURE: NA

DATE OF PRESENT RANK OR TITLE: 2026

EDUCATION:

2011. Ph.D. Ecology and Evolutionary Biology, minor in Statistics. Iowa State University. Ames, IA.
2005. M.S. Forest Science. Universidad de Chile. Santiago, Chile.
2002. M.E.M. Environmental Management. Portland State University. Portland, OR.
1997. B.S. Biological Sciences. University of California. Davis, CA.

EXPERIENCE:

Teaching, Extension and Research Appointments:

2026- Present: Associate Professor of Biometrics. University of Idaho. Moscow, ID.
2021-2026: Assistant Professor of Biometrics. University of Idaho. Moscow, ID.
2021- Present: Assistant Unit Leader. USGS, Idaho Cooperative Fish and Wildlife Research Unit. Moscow, ID.
2009-2010: Instructor of Evolution. Oregon State University. Corvallis, OR.
2008: Instructor of Cell Biology. Mt. Hood Community College. Gresham, OR.
2003-2005: Fellow. Organization of American States. Santiago, Chile.

Non-Academic Employment:

2010-2021: Quantitative Fish Biologist. Oregon Department of Fish and Wildlife.

1) Deliver scientific and technical support to implement conservation and fishery management goals.

- Served on leadership team that developed Coastal Multispecies Conservation Plan.
- Assembled data, developed appropriate models, and reported metrics of population health for the Rogue Fall Chinook Conservation Plan and the Coastal Multispecies Conservation Plan, and Coastal Coho Recovery Plan, and the Rogue and South Coast Multispecies Plan.
- Answered questions from staff, stakeholders, and technical reviewers about conservation plan analyses.
- Modeled abundance metrics needed to implement coastal coho fishery.
- Developed methods to calculate the proportion of hatchery fish spawning naturally.

2) Represent agency and communicate science to diverse audiences.

- Member of the Hatchery Scientific Review Group, established by the U.S. Congress.
- Taught workshops and delivered presentations at professional societies and academic institutions
- Published ideas and original research in prestigious international journals and subject-specific journals.
- Provided peer review to authors submitting articles to leading academic journals.
- Spoke at international and local professional society meetings, university, etc.

3) Create complex mathematical, statistical, and ecological simulation models to answer natural resource management questions.

- Developed hierarchical, state-space Bayesian statistical stock-recruitment models to estimate magnitude of uncertainty in estimates of fish population trend, productivity, carrying capacity, and abundance-habitat relationships.
 - Programmed numerous population viability models that use Bayesian estimates of uncertainty.
 - Developed novel model to synthesize the genetic and demographic interactions of hatchery and wild fish.
 - Conducted analyses to determine effects of harvest on population viability.
 - Developed salmon forecasting techniques using artificial intelligence.
 - Added a time domain to existing spatial mark-recapture models to model habitat use.
- 4) Develop decision support tools.
- Created Bayesian belief network models to capture the structure of decisions about research priorities, tide gates restoration, cold water refugia, and hatchery practices.
 - Developed rule-set guiding the harvest bag limits of coastal fall Chinook.
 - Optimized effort across multi-objective monitoring using calculus and stochastic dynamic programming.
 - Created geospatial tool for designing biological reserves on stream networks for statewide habitat prioritization.
- 5) Provide statistical consulting for staff at ODFW.
- Implemented regression, ANOVA, BACI, generalized linear models, mixed effects models, hierarchical models, meta-analysis, generalized additive models, variograms, kriging, conditional autoregressive spatial analysis, time series analysis, state-space models, PCA, MANOVA, ordination, regression trees, neural networks, LASSO and other shrinkage techniques, model selection with AIC, BIC, WAIC, DIC, Bayes factors, likelihood ratios.
 - Frequently addressed questions from staff about power analysis, mark-recapture analysis, monitoring designs, spatially-explicit analyses, analysis of existing data.
 - Provided review and recommendations for economic/budget analyses for executive administrators.

1998-2000: Volunteer. US Peace Corps, Nicaragua.

- Developed environmental education curricula for Parque Nacional Volcán Masaya.
- Taught English as a second language in the community of Nindirí, Nicaragua.
- Formally scored the highest rate of Spanish improvement during 2 years among my cohort of 20 volunteers.

Honors and Awards:

- *Team Pride Award*, Oregon Department of Fish and Wildlife, 2014. Out of ~1000 FTEs.
- *Award of Excellence*, Oregon Department of Fish and Wildlife, 2015. Out of ~ 600 FTEs.
- *Individual Pride Award*, Oregon Department of Fish and Wildlife, 2016. Out of ~ 1000 FTEs.

TEACHING ACCOMPLISHMENTS:

Areas of Specialization: Population Ecology, Statistics, Decision Analysis.

Courses Taught

- Ecological Modeling. University of Idaho. WLF 552. Fall 2021, Fall 2023.
- Bayesian Analysis in Ecology. University of Idaho. WLF 558. Spring 2023, Spring 2025.
- Evolution. Oregon State University. BI445/545. Winter 2009, Spring 2010.
- Cell Biology. Mt. Hood Community College. BI101. Fall 2008.

Students Advised:

Graduate students:

- Ryan Vosbigian. M.S. Fisheries Resources. Graduated Summer 2024. Now *Fisheries Biometrician* at Quinault Indian Nation.
- Briana Lubenau. M.S. Wildlife Resources. Co-advised. Graduated Fall 2024. Now *Biologist* at USFWS.
- Joshua Heishman. M.S. Fisheries Resources. Graduated Spring 2025. Now *Stock Assessment Biologist* at Upper Skagit Indian Tribe.

- Samuel Foster. Ph.D. Wildlife Resources. Graduated Spring 2025. Now *Wildlife Biologist* for Ministry in British Columbia.
- Cameron Macias. Ph.D. Wildlife Resources. Co-advised. Graduated Summer 2025. Now *Wildlife Biologist* for Lower Elwha Klallam Tribe.
- Nathan Nadal. M.S. Fisheries Resources. In progress.
- Manuel Carballo. M.S. Fisheries Resources. In progress.
- Jacob Russell. Ph.D. Environmental Sciences. Co-advised. In progress.

Post-doctoral scholars

- Sarah Bassing. Spring 2025. Now *Assistant Professor*, Montana State University.

Paid internships supervised

- Tamsen Farris. Summer 2025-Spring 2026. Undergraduate in Modeling and Data Science working on steelhead ocean survival.
- Logan Wendling. Summer 2021. M.S. Statistics. Now *Fisheries Biometrician*, Alaska Department of Fish and Game.
- Jacob Hofer. Summer 2024. M.S. Statistics. Now *Statistician I*, Northrop Grumman Aeronautics.
- Jo Ash. Summer 2024 – present. Graduate of MNR program.

Service on graduate committees

- Donovan Maude. M.S. Fisheries Resources. Spring 2024.
- Megan Wright/Turnock. M.S. Wildlife Resources. Spring 2024.
- Katrina Lopez. M.S. Wildlife Resources. Spring 2024.
- Katie Huggler. Ph.D. Wildlife Resources. Spring 2024.
- Lilian Harrison. M.S. Wildlife Resources. Fall 2024.
- Nick Voss. Ph.D. Fisheries Resources. Summer 2025.
- Julia Rushton. M.S. Agriculture. Summer 2025.
- Ariana Cerreta. Ph.D. Wildlife Resources. Spring 2026.
- Brent Barry. Ph.D. Wildlife Resources. In progress.
- Garrison Ferone. Ph.D. Wildlife Resources. In progress.
- Fiona McKibben. Ph.D. Wildlife Resources. In progress.
- David Arthaud. Ph.D. Fisheries Resources. Inactive.

Courses Developed:

- Ecological Modeling. University of Idaho. WLF 5520.
- Bayesian Analysis in Ecology. University of Idaho. WLF 5580.
- Evolution. Oregon State University. BI445/545.

Non-credit Classes, Workshops, Seminars, Invited Lectures, etc.:

- Falcy, M.R.** 2026. The Ecology of Uncertainty: Models, Decisions, and Management. Invited seminar. University of Washington.
- Falcy, M.R.** 2026. The Ecology of Uncertainty: Models, Decisions, and Management. Invited seminar. Department of Fish and Wildlife Sciences, University of Idaho.
- Falcy, M.R.** 2026. Tools for career success in a changing hiring landscape. Workshop at the Annual Meeting of the Idaho Chapter of the American Fisheries Society.
- Falcy, M.R.** 2026. Decision Making in Fish and Wildlife Management. Guest lecture in FISH5100 – Advanced Fish and Wildlife Management. University of Idaho.
- Falcy, M.R.** 2025. Applications of Quantitative Analysis in Ecology. Invited seminar, University of Northern British Columbia.
- Falcy, M.R.** 2025. Analytical Ecology. Invited seminar. Department of Fish and Wildlife Sciences, University of Idaho.
- Falcy, M.R.** 2025. Paradigms of statistical inference. Guest lecture in WLF 550- Quantitative Analysis of Fish and Wildlife Populations. University of Idaho.

- Falcy, M.R.** 2024. Ecological Application of Statistical Theory. Invited seminar. Department of Mathematics and Statistics, University of Idaho.
- Falcy, M.R.** 2024. Analytical Ecology. Invited seminar. Department of Mathematics and Statistical Science, University of Idaho.
- Falcy, M.R.** 2024. Decision Making in Fish and Wildlife Management. Guest lecture in FISH510 – Advanced Fish and Wildlife Management. University of Idaho.
- Falcy, M.R.** 2024. Introduction to Bayesian Analysis. 8-hour workshop at the annual meeting of the Idaho Chapter of the Wildlife Society. Coeur d’Alene, ID.
- Falcy, M.R.** and Scheuerell, M. 2024. Salmon Forecasting Workshop. Annual meeting of the American Fishery Society of ID, WA, BC. Spokane, WA.
- Peterson, J., **Falcy, M.R.**, Scheuerell, M., and Som, N. 2024. Cooperative Fish and Wildlife Research Units Cooperator Run Forecasting Workshop. 16-hour event hosted by US Geological Survey. Wilsonville, OR.
- Falcy, M.R.** 2023. Paradigms of statistical inference. Guest lecture in WLF 550- Quantitative Analysis of Fish and Wildlife Populations. University of Idaho.
- Falcy, M.R.** 2023. Theory and practice of scientific decision-making. Guest lecture in WLF 492 – Wildlife Management.
- Falcy, M.R.** 2022. Scientific decision-making. Guest lecture in WLF 492 – Wildlife Management.
- Falcy, M.R.** 2022. Decision Making in Fish and Wildlife Management. Guest lecture in FISH510 – Advanced Fish and Wildlife Management. University of Idaho.
- Falcy, M.R.** 2021. Paradigms of statistical inference. Guest lecture in WLF 550- Quantitative Analysis of Fish and Wildlife Populations. University of Idaho.
- Falcy, M.R.** 2019. SNO BiRD: Stream network optimizer for biological reserve design. Invited seminar. Fisheries Department, Oregon State University. Corvallis, OR.
- Falcy, M.R.** 2019. BioRogy. Four-week review of R programming and statistical inference for Oregon Department of Fish and Wildlife professionals.
- Falcy, M.R.** 2018. Effects of sea lions on Willamette River winter steelhead viability. Invited seminar. Fisheries Department, Oregon State University. Corvallis, OR.
- Falcy, M.R.** 2018. Quantitative and Computational Ecology. 8-hour seminar in the Department of Fisheries and Wildlife, Oregon State University. Corvallis, OR.
- Falcy, M.R.** 2017. Conservation decision making: Integrating the precautionary principle with uncertainty. Invited seminar. Fisheries Department, Oregon State University. Corvallis, OR.
- Falcy, M.R.** 2017. Bayesian Analysis for Beginners. 2-day workshop at the annual meeting of the Oregon Chapter of the American Fisheries Society. Bend, OR.

SCHOLARSHIP ACCOMPLISHMENTS:

Peer Reviewed Papers

* = advisee. Papers listed as “in prep” have been fully written but are undergoing refinement prior to submission.

- Vosbigian*, R. A., M. Dobos, and M.R. **Falcy**. 2026. An age- and time-specific space-for-time mark-recapture model. Target journal: *Methods in Ecology and Evolution* (In prep).
- Heishman*, J., S. Wilson, A. Black, N. Jensen, and M.R. **Falcy**. 2026. Beyond put and take: A quantitative framework for population-level decision-making. Target journal: *Fisheries* (In prep).
- Vosbigian*, R., D.W. Whitney, J.M. DuPont, and M.R. **Falcy**. 2026. Evaluating the influence of bag and size limits on harvest of steelhead. Target journal: *North American Journal of Fisheries Management* (In prep).
- Macias*, C.L., M.R. **Falcy**, T. Levi, L.M. Elbroch, K.A. Sager-Fradkin, and L.P. Waits. 2026. Prey composition and dietary overlap of bobcats and cougars using GPS collar data and DNA metabarcoding in western Washington State. Target journal: *Environmental DNA* (In prep).
- Macias*, C.L., M.R. **Falcy**, B.C. Augustine, J.R. Adams, S. Cendejas-Zarelli, S.H. Williams, L.M. Elbroch, K.A. Sager-Fradkin, and L.P. Waits. 2026. Evaluating new abundance estimators in real-world applications: Cameras and DNA yield comparable density estimates for cryptic carnivores in western Washington State. Target journal: *Journal of Wildlife Management* (In prep).
- Macias*, C.L., M.R. **Falcy**, J.R. Adams, L.M. Elbroch, K.A. Sager-Fradkin, and L.P. Waits. 2026. Evaluating fecal DNA preservation and extraction methods for noninvasive wildlife genetic sampling. Target journal:

- Conservation Genetics Resources (In prep).
- Foster*, S., A. Ford, and M.R. **Falcy**. 2026. Community reordering along disturbance gradients: Roads filter mammal composition. Target journal: Canadian Journal of Zoology (In prep).
- Foster*, S., A. Ford, and M.R. **Falcy**. 2026. Seasonal and bottom-up processes drive weekly habitat use of mule deer more than perceived predation risks. Target journal: Journal of Wildlife Management (In prep).
- Foster*, S., A. Ford, and M.R. **Falcy**. 2026. Caught in a time trap: Animal activity in the Anthropocene. Target journal: Frontiers in Ecology and the Environment (In prep).
- Harrison, L.A., K.S. Christie, J.L. Rachlow, and M.R. **Falcy**. 2026. Pika behavioral response to climate during the summer foraging season at high latitudes. Target journal: Journal of Mammalogy (In prep).
- Ash*, J., C. Caudill, and M.R. **Falcy**. 2026. The evolution of philopatry on spatially structured and temporally trending landscapes. Target journal: Ecography (In prep).
- Farris*, T., R.A. Vosbigian*, T. Copeland, M. Dobos, and M.R. **Falcy**. 2026. Snake River steelhead ocean survival driven by early ocean conditions and smolt size. (In prep).
- Russell*, J., R.A. Vosbigian*, and M.R. **Falcy**. 2026. Misinformed by non-informative priors: Jeffreys' method ensures consistent Bayesian inference. Ecology (In review).
- Falcy**, M.R. 2026. Data-driven decision-making with novel probability density functions. Methods in Ecology and Evolution (In review).
- Falcy**, M.R., and C.M. Lorion. 2026. Decision-making processes. In *Inland Fisheries Management in North America*, 4th ed. (In review).
- Heishman*, J., A. Black, S. Wilson, and M.R. **Falcy**. 2026. An open-population demographic analysis of burbot in the Kootenai River to inform fishery management. Transactions of the American Fisheries Society (In review).
- Mumma, M., E. Roche, Z. Fogel, J. Tebbenkamp, J. Struthers, M. Parsons, K. Oelrich, M. Campbell, M.R. **Falcy**, and S. Roberts. 2026. Estimating gray wolf (*Canis lupus*) abundance using close-kin data and approximate Bayesian computation. Ecological Applications (In review).
- Sabal, M.C., E. Leonetti, M.R. **Falcy**, P. Stevens, J. Anthony, and C. Lorion. 2026. Forecasting spring Chinook salmon abundances amid a changing climate on the Umpqua River, Oregon. Canadian Journal of Fisheries and Aquatic Sciences (In revision).
- Bassing*, S.B., D.E. Ausband, M.A. Mumma, J. Baumgardt, S. Thompson, M. Hurley, and M.R. **Falcy**. 2026. Disentangling the web of species interactions in a multi-predator, multi-prey community. Ecology (In revision).
- Vosbigian*, R., J.H. Powell, and M.R. **Falcy**. 2026. Integrating data on a mixed stock anadromous fishery to estimate stock-specific escapement and mortalities. North American Journal of Fisheries Management 46:447-462.
- Falcy**, M.R. 2026. Refining indices of risk from salmonid hatcheries. North American Journal of Fisheries Management 46:1-8.
- Vosbigian*, R., A. Ballinger, T.E. Link, T. Copeland, and M.R. **Falcy**. 2025. Elevation mediates juvenile steelhead demographic response to stream temperature and flow. Canadian Journal of Fisheries and Aquatic Sciences 82: 1-16.
- Harrison, L.A., K.S. Christie, C. Brandt, M.R. **Falcy**, R.A. Long, S.L. Gilbert, and J.L. Rachlow. 2025. Mechanism influencing thermal refuges and territory occupancy by pikas during summer and winter. Arctic, Antarctic, and Alpine Research 57, No.1, 2502161.
- Bassing*, S.B., D.E. Ausband, M.A. Mumma, S. Thompson, M.A. Hurley, and M.R. **Falcy**. 2025. Mammalian predator co-occurrence affected by prey and habitat more than competitor presence at multiple time scales. Ecological Monographs 95:e164: 1-25.
- Vosbigian*, R., L. Wendling*, T. Copeland, and M.R. **Falcy**. 2024. Cycles in adult steelhead length suggest interspecific competition in the North Pacific Ocean. Canadian Journal of Fisheries and Aquatic Sciences 81: 1666-1675.
- Falcy**, M.R. and R.J. Constable Jr. 2024. Quantifying uncertainty when extrapolating the relationship between snorkel counts and mark-recapture estimates of juvenile salmonids. Canadian Journal of Fisheries and Aquatic Sciences 81: 1279-1291.
- Anlauf-Dunn, K.J., B.J. Clemens, M.R. **Falcy**, and C. Zambory. 2024. Spatio-temporal distribution of adult Pacific lamprey *Entosphenus tridentatus* relative to habitat fragmentation. River Research Applications 1-15.
- Turnock, M.F., L.P. Waits, J.E. Teisberg, W.F. Kasworm, M.R. **Falcy**, and M.F. Proctor. 2024. Gene flow prevents genetic diversity loss despite small effective population size in fragmented grizzly bear (*Ursus arctos*) populations. Conservation Genetics 26: 279-291.

- Johnson, M.A., M.K. Jones, M.R. **Falcy**, J. Spangler, R.B. Couture, and D.L.G. Noakes. 2023. Can angler-assisted broodstock collection programs improve harvest rates of hatchery-produced steelhead? *Environmental Biology of Fishes* 106: 1079-1092.
- Falcy**, M.R. 2023. Inferring hatchery effects using spawner-recruit data: Comment on Courter et al. (2022). *Canadian Journal of Fisheries and Aquatic Sciences*. 80: 420-421.
- Falcy**, M.R. 2021. Using social values in the prioritization of research: Quantitative examples and generalizations. *Ecology and Evolution* 11: 18000-18010.
- Falcy**, M.R. 2018. A cost-optimization framework for planning applied environmental science. *BioScience* 68:912-922
- Falcy**, M.R. and E. Suring. 2018. Detecting the effects of management regime shifts in dynamic environments using multi-population state-space models. *Biological Conservation* 221:34-43.
- Falcy**, M.R. 2016. Conservation decision making: Integrating the precautionary principle with uncertainty. *Frontiers in Ecology and the Environment* 14:499-504.
- Falcy**, M.R., J.L. McCormick, and S.A. Miller. 2016. Proxies in practice: Calibration and validation of multiple indices of animal abundance. *Journal of Fish and Wildlife Management*. 7: 117-128.
- Falcy**, M.R. 2015. Righting the resource management ship: a response to Grose. *Trends in Ecology and Evolution* 30:294-295.
- Falcy**, M.R. 2015. Density-dependent habitat selection of spawning Chinook salmon: broad scale evidence and implications. *Journal of Animal Ecology* 84: 545-553.
- McCormick, J.L., and M.R. **Falcy**. 2015. Evaluation of non-traditional modelling techniques for forecasting salmon returns. *Fisheries Management and Ecology* 22:269-278.
- Falcy**, M.R., and B.J. Danielson. 2014. Post-hurricane recovery and long-term viability of the Alabama beach mouse. *Biological Conservation* 178: 28-36.
- Falcy**, M.R., and B.J. Danielson. 2013. A complex relationship between moonlight and temperature on the foraging behavior of the Alabama beach mouse. *Ecology* 94: 2632-2637.
- Falcy**, M.R., and B.J. Danielson. 2013. Assessment of competitive release of endangered beach mouse (*Peromyscus polionotus ammobates*). *Journal of Mammalogy* 94: 584-590.
- Duthie, A.B., and M.R. **Falcy**. 2013. The influence of habitat autocorrelation on plants and their seed-eating pollinators. *Ecological Modelling* 251:260-270.
- Bangs, B., M.R. **Falcy**, P.D. Scheerer, and S. Clements. 2013. Comparison of three methods for marking a small floodplain minnow. *Animal Biotelemetry* 1:18.
- Falcy**, M.R., and B.J. Danielson. 2011. When sinks rescue sources in dynamic environments. In Liu, J., Hull, V., Morzillo, A., and Weins, J. (Eds.). *Sources, Sinks, and Sustainability*. Cambridge Studies in Landscape Ecology.
- Chilcote, M.W., K.W. Goodson, and M.R. **Falcy**. 2011. Reduced recruitment performance in natural populations of anadromous salmonids associated with hatchery-reared fish. *Canadian Journal of Fisheries and Aquatic Sciences* 68:511-522.
- Deitloff, J., M.R. **Falcy**, J.D. Krenz, and B.R. McMillan. 2010. Correlating small mammal abundance to climatic variation over twenty years. *Journal of Mammalogy* 91(1):193-199.
- Falcy**, M.R. 2009. Theoretical perspectives on habitat destruction. In Kudrow, N.J. (Ed.). *Conservation of Natural Resources*. Nova Sciences Publishers. ISBN: 978-1-60741-178-9
- Falcy**, M.R., and C.F. Estades. 2007. Effectiveness of corridors relative to enlargement of habitat patches. *Conservation Biology* 21(5):1341-1346.

Management Reports

- i. Falcy is solo author
- ii. Falcy is lead author
- iii. Falcy is a contributing author

Rogue-South Coast Multi-Species Conservation and Management Plan.ⁱⁱⁱ 2021. Oregon Department of Fish and Wildlife.

Elk River Fall Chinook: Status Assessment Update.ⁱ 2020. Oregon Department of Fish and Wildlife.

Coastal Fall Chinook Sliding Scale Bag Limits: A proposal for pre-season inland harvest management.ⁱ 2019. Oregon Department of Fish and Wildlife.

Estimating the weighted proportion of hatchery-origin spawners, pHOSw.ⁱ 2019. Oregon Department of Fish and Wildlife.

Rogue Spring Chinook Salmon Conservation Plan Comprehensive Assessment and Update.ⁱⁱⁱ 2019. Oregon

Department of Fish and Wildlife
 Spring Chinook Status Assessment.ⁱ 2018. Oregon Department of Fish and Wildlife.
 Population Viability of Willamette River Winter Steelhead: An Assessment of the Effects of Sea Lions at Willamette Falls.ⁱ 2017. Oregon Department of Fish and Wildlife.
 Habitat Use by Bluegills *Pomoxis annularis* and Oregon Chub *Oregonichthys crameri* in an Off-channel Habitat in the Willamette Valley.ⁱⁱⁱ 2017. Oregon Department of Fish and Wildlife.
 Power analysis: determination of sub-sample size for analysis of fish scales from coastal Chinook salmon from spawning ground surveys.ⁱⁱ 2014. Oregon Department of Fish and Wildlife.
 Coastal Multi-Species Conservation and Management Plan.ⁱⁱⁱ 2014. Oregon Department of Fish and Wildlife.

Professional Meeting Papers:

- Falcy, M.R.** 2026. Organizer and speaker, invited symposium: Using uncertainty in decision-making. Annual Meeting of the Ecological Society of America. Salt Lake City, UT.
- Falcy, M.R.** 2026. Data-driven decision making with novel probability density functions. Annual Meeting of the Idaho Chapter of The Wildlife Society. Fort Hall, ID.
- Falcy, M.R.** 2026. Refining indices of risk from salmonid hatcheries. Annual Meeting of the Idaho Chapter of the American Fisheries Society. Idaho Falls, ID.
- Falcy, M.R.** 2024. A dynamic eradication strategy. 154th Annual Meeting of the American Fisheries Society. Honolulu, HI.
- Falcy, M.R.** 2022. Quantifying the relationship between snorkel counts and mark-recapture estimates of juvenile salmonids at multiple scales. 152nd Annual Meeting of the American Fisheries Society. Spokane, WA.
- Falcy, M.R.** 2021. Combining matrix models and decision theory to manage steelhead harvest. Pacific Coast Steelhead Management Meeting (Online).
- Falcy, M.R.** 2020. Science as Bayesian model averaging, with applications to density dependence. Annual meeting of the Oregon Chapter of the American Fisheries Society. Bend, OR.
- Falcy, M.R.** 2019. Middle Fork John Day Spring Chinook life cycle PVA. AMIP LCM Meeting. Hood River, OR.
- Falcy, M.R.** 2019. Detecting management regime shifts in dynamic environments. North Pacific Anadromous Fish Commission Meeting. Portland, OR.
- Falcy, M.R.** 2019. Designing biological reserves on stream networks. Annual meeting of the Oregon Chapter of American Fisheries Society. Bend, OR.
- Falcy, M.R.** 2018. Effects of sea lion predation on Willamette River winter steelhead viability. Salmon Ocean Ecology Meeting. Newport, OR.
- Falcy, M.R.** 2018. Effects of sea lions on Willamette River winter steelhead viability. U.S. Army Corps of Engineers Willamette River Science Review. Corvallis, OR.
- Falcy, M.R.** 2018. Population viability of Willamette River winter steelhead: An assessment of the effects of sea lions at Willamette Falls. Pacific Coast Steelhead Management Meeting. WallaWalla, Wa.
- Falcy, M.R.** 2017. Detecting the effects of management regime shifts in dynamic environments using multi-population state-space models. National Meeting of the Ecological Society of America. Portland, OR.
- Falcy, M.R.** 2016. Conservation decision making: Integrating the precautionary principle with uncertainty. Annual meeting of the Oregon Chapter of the American Fisheries Society. Seaside, OR.
- Falcy, M.R.** 2015. Combining genetics and demographics in a viability model of hatchery-wild systems subject to environmental change. National Meeting of the American Fisheries Society. Portland, OR.
- Falcy, M.R.** 2014. Estimating spawner population abundance of coastal Fall Chinook: is non-random sampling optimal? Annual meeting of the Oregon Chapter of the American Fisheries Society. Eugene, OR.
- Falcy, M.R.** 2014. Exploring patterns in coastal coho abundance and productivity. Annual meeting of the Oregon Chapter of the American Fisheries Society. Eugene, OR.
- Falcy, M.R.** 2013. Population viability analysis with Bayesian state-space parameter estimation. The International Environmetrics Society. Anchorage, AK.
- Falcy, M.R.** 2013. A Bayesian meta-analytic assessment of the effect of watershed geomorphology and lithology on chinook salmon recruitment. Annual meeting of the Oregon Chapter American Fisheries Society. Bend, OR.
- Falcy, M.R.** 2012. Artificial neural networks: finding complex environmental signals or just fitting noise? Oregon Chapter of the American Fisheries Society- Forecasting Symposium. Portland, OR.

- Falcy, M.R.** 2011. Individual and population-level responses of the Alabama beach mouse (*Peromyscus polionotus ammobates*) to environmental variation in space and time. Biannual beach mouse conference. Daphne, AL.
- Falcy, M.R.** 2010. Post-hurricane recovery and viability of the Alabama beach mouse. Annual meeting of the American Society of Mammalogists. Laramie, WY.
- Falcy, M.R.** 2009. Optimal habitat selection of predator and prey in explicit space. Iowa State University Ecology and Evolutionary Biology Spring Symposium. Ames, IA.

Co-authored Presentations (presented by advisees or collaborators; * = advisee):

- Bassing*, S.B., D.E. Ausband, and M.R. **Falcy**. 2026. Embracing uncertainty: Bayesian belief networks as a tool for a reintroduction decision. Annual Meeting of the Ecological Society of America. Salt Lake City, UT.
- Vosbigian*, R., M. Dobos, and M.R. **Falcy**. 2025. Modeling freshwater survival of wild juvenile steelhead. Pacific Coast Steelhead Management Meeting. Boise, ID.
- Vosbigian*, R., M. Dobos, and M.R. **Falcy**. 2025. Evaluating juvenile life history strategies in Snake River steelhead to inform effective habitat restoration. Annual Meeting of the Idaho Chapter of the American Fisheries Society. Boise, ID.
- Bassing*, S.B., D.E. Ausband, M.A. Mumma, J. Baumgardt, S. Thompson, M.A. Hurley, and M.R. **Falcy**. 2025. Disentangling the web of species interactions in northern Idaho's large mammal community. Annual Meeting of the Idaho Chapter of The Wildlife Society. Moscow, ID.
- Vosbigian*, R., A. Ballinger, T. Copeland, and M.R. **Falcy**. 2024. Effects of climate change on juvenile steelhead abundance and size structure. 154th Annual Meeting of the American Fisheries Society. Honolulu, HI.
- Verity, H., and M.R. **Falcy**. 2024. Sifting snow: Foraging tradeoffs of snowshoe hares on Moscow Mountain, Idaho. Annual Meeting of the Idaho Chapter of The Wildlife Society. Coeur d'Alene, ID.
- Vosbigian*, R., J.H. Powell, and M.R. **Falcy**. 2024. Integrating data on the Snake River steelhead fishery enhances reporting of protected natural-origin stocks. Joint Meeting of the Washington-British Columbia and Idaho Chapters of the American Fisheries Society. Spokane, WA.
- Heishman*, J., A. Black, S. Wilson, and M.R. **Falcy**. 2024. Abundance and density dependent survival of burbot in the Kootenai River's hatchery-augmented fishery. Joint Meeting of the Washington-British Columbia and Idaho Chapters of the American Fisheries Society. Spokane, WA.
- Lubenau*, B.M., D.E. Ausband, M.R. **Falcy**, et al. 2024. New boot scootin': The migratory plasticity of southern mountain caribou. Annual Meeting of the Idaho Chapter of The Wildlife Society. Coeur d'Alene, ID. Also presented at U.S. Fish and Wildlife Service Science of the Service, Idaho.
- Bassing*, S.B., D.E. Ausband, M.A. Mumma, M.A. Hurley, S. Thompson, and M.R. **Falcy**. 2024. Predator co-occurrence affected by prey and habitat more than competitor presence at multiple time scales. Annual Meeting of the Idaho Chapter of The Wildlife Society. Coeur d'Alene, ID.
- Heishman*, J., and M.R. **Falcy**. 2023. Age-structured survival rates of burbot in the Kootenai River's hatchery-augmented fishery. Annual Meeting of the Idaho Chapter of the American Fisheries Society. Boise, ID.
- Vosbigian*, R., L. Wendling*, T. Copeland, and M.R. **Falcy**. 2023. Two-year cycles in Snake River steelhead growth suggest competition with pink salmon in the Pacific Ocean. Annual Meeting of the Idaho Chapter of the American Fisheries Society. Boise, ID.
- Winkel, B., D.E. Ausband, M.R. **Falcy**, et al. 2023. Migratory patterns and plasticity of southern mountain caribou in a changing landscape. North American Caribou Workshop and Arctic Ungulate Conference. Anchorage, AK.
- Bassing*, S.B., D.E. Ausband, M.A. Mumma, and M.R. **Falcy**. 2023. Spatial patterns of predator interactions in northern Idaho: Preliminary findings. Annual Conference of the Idaho Chapter of The Wildlife Society. Boise, ID.

Grants and Contracts Awarded: (provide principal and co investigators, title, sponsor, funding dates, amount)

- Ausband, D.A., and M.R. **Falcy** (Co-PI). Caribou Recovery in Northern Idaho. US Fish and Wildlife Service. August 2021- May 2025. \$330,000.
- Falcy, M.R.** Snake River Steelhead Run Reconstruction. Idaho Department of Fish and Game. May 2022-August 2024. \$104,000
- Falcy, M.R.** Burbot Population Dynamics in Kootenai River. Idaho Department of Fish and Game. August

2022- May 2025. \$100,723

Falcy, M.R. and Ausband, D. (Co-PI). Patterns and Processes in Multi-predator Dynamical Systems. Idaho Department of Fish and Game. August 2023-December 2024. \$180,283.

Falcy, M.R. British Columbia Mule Deer Habitat and Communities. MITACS Accelerate. \$60,609

Falcy, M.R. Mammal Predator Occurrence Patterns in British Columbia. HCTF. \$10,000

Falcy, M.R. Ecotoxicological Inference in the Kootenai River, Idaho. Kootenai Tribe of Idaho. August 2024-May 2026. \$150,950.

Falcy, M.R. Mountain Whitefish Diet Analysis. Kootenai Tribe of Idaho. August 2024-May 2026. \$140,484.

Link, T. and M.R. Falcy (Co-PI). Linking Dynamic Microclimatic Refugia to Habitat Use Strategies. USGS, NW Climate Adaptation Science Center. October 2024 – December 2027. \$329,597.

SERVICE:

Committee Assignments:

- Scholarship Committee. Department of Fish and Wildlife Sciences. 2022-present
- Internship Committee. Department of Fish and Wildlife Sciences. 2022-present

Professional and Scholarly Organizations

- Member: American Fisheries Society
- Member: The Wildlife Society

Outreach Service:

- As a federal (US Geological Survey) employee, I am discouraged from speaking with the press.

Community Service: (non-academic unrelated to employment)

- Led learning activity for 12-16 year-olds about the role of math in fish and wildlife management at the University of Idaho's Natural Resources Camp. Ketchum Idaho. June 2024.

PROFESSIONAL DEVELOPMENT:

Administration/Management:

Foundations of Leadership. 56-hour course offered by the State of Oregon.